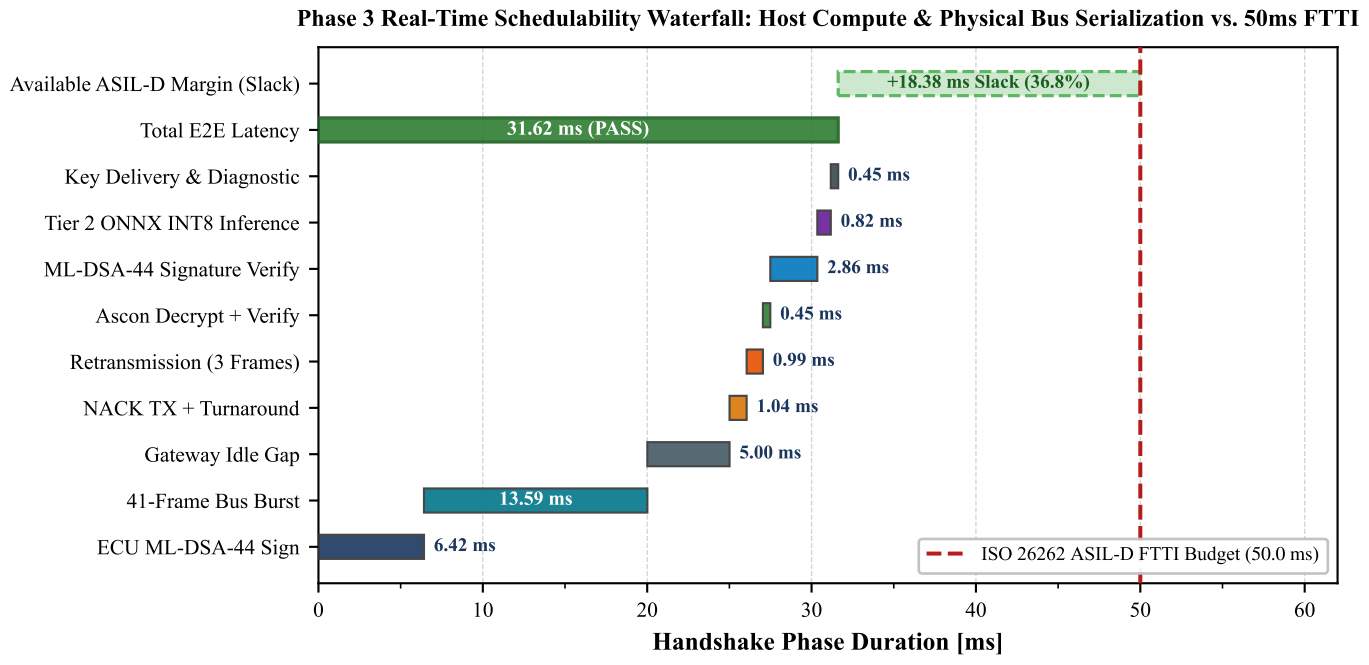
Phase 3 Real-Time Schedulability Waterfall: Host Compute & Physical Bus Serialization vs. 50ms FTTI
Available ASIL-D Margin (Slack)
Total E2E Latency
Key Delivery & Diagnostic
Tier 2 ONNX INT8 Inference
ML-DSA-44 Signature Verify
Ascon Decrypt + Verify
Retransmission (3 Frames)
NACK TX + Turnaround
Gateway Idle Gap
41-Frame Bus Burst
ECU ML-DSA-44 Sign
+18.38 ms Slack (36.8%)
31.62 ms (PASS)
0.45 ms
0.82 ms
2.86 ms
0.45 ms
0.99 ms
1.04 ms
5.00 ms
13.59 ms
6.42 ms
ISO 26262 ASIL-D FTTI Budget (50.0 ms)
0
10
20
30
40
50
60
Handshake Phase Duration [ms]